

Documentation(Screenshots) of testing ReadFile and ParticleProperties

```
time,particles,data = Read('MW_000-Copy1.txt')
print(data[13])
```

```
(1., 0.00394985, 192.281, 9.75553, -92.0792, 58.045, -13.1158, -20.966)
```

```
Distance_Mag = np.around((np.sqrt((x**2 + y**2 + z**2)/u.kpc),3)

Velocity_Mag = np.around((np.sqrt((vx**2 + vy**2 + vz**2))*u.km/u.s),3)

Mass = (mass*10**10)*u.M_sun

return Distance_Mag, Velocity_Mag, Mass
```

```
ParticleInfo('MW_000-Copy1.txt',2,3)
```

```
z is 75000
```

```
g is 135000
```

```
(<Quantity 6.729 kpc>, <Quantity 241.415 km / s>, <Quantity 1000000. solMass>)
```

```
print(particles)
#print(data)

return time,particles,data #returns the desired outputs

time,particles,data = Read('MW_000-Copy1.txt')
print(data['x'][2])
```

```
-0.505945
```

```
file = open(filename, 'r') #Opens the file that we read in as filename

line1 = file.readline() #Reads the first line of the file and makes it into a string
label, value = line1.split() #Splits the string containing first line of file into a label and a value (time, actual time in a number)
time = float(value)*u.Myr #converts the value in numbers from the file into a floating point value and puts it in units of Myr
                        #(gives us the time of the file)

line2 = file.readline() #Reads second line of file and makes it into a string
label, value = line2.split() #Splits the string containing second line of file into a label and a value (time, actual time in a number)
particles = float(value) #converts the value in numbers from the file into a floating point value(gives us total particles in file)

file.close() #Closes file

data = np.genfromtxt(filename, dtype=None, names=True, skip_header=3) #Creates an array to hold the rest of the data from the file
#print(time)
#print(particles)
#print(data)

return time,particles,data #returns the desired outputs

time,particles,data = Read('MW_000-Copy1.txt')
print(data['type'][2])
```

```
1.0
```

```

    return Distance_Mag, Velocity_Mag, Mass

#ParticleInfo('MW_000-Copy1.txt',2,99)
Distance_Mag, Velocity_Mag, Mass = ParticleInfo('MW_000-Copy1.txt',2,99)
print(Distance_Mag.to(u.lyr))

13845.338234075754 lyr

```

```

ParticleInfo('MW_000-Copy1.txt',2,99)

(<Quantity 4.245 kpc>, <Quantity 312.135 km / s>, <Quantity 1000000. solMass>)

```

```

Distance_Mag = np.around((np.sqrt((x**2 + y**2 + z**2))*u.kpc),3)

Velocity_Mag = np.around((np.sqrt((vx**2 + vy**2 + vz**2))*u.km/u.s),3)

Mass = (mass*10**10)*u.M_sun

    return Distance_Mag, Velocity_Mag, Mass

ParticleInfo('MW_000-Copy1.txt',3,15)

```

```

l4]: (<Quantity 4.282 kpc>, <Quantity 507.466 km / s>, <Quantity 1000500. solMass>)

```

```

    return Distance_Mag, Velocity_Mag, Mass

#ParticleInfo('MW_000-Copy1.txt',1,1)
Distance_Mag, Velocity_Mag, Mass = ParticleInfo('MW_000-Copy1.txt',2,99)
print(np.around(Distance_Mag.to(u.lyr),3))

13845.338 lyr

```

```

:
```

```

:
```

```

Distance_Mag = np.around((np.sqrt((x**2 + y**2 + z**2))*u.kpc

Velocity_Mag = np.around((np.sqrt((vx**2 + vy**2 + vz**2))*u.

Mass = (mass*10**10)*u.M_sun

    return Distance_Mag, Velocity_Mag, Mass

articleInfo('MW_000-Copy1.txt',1,2)

```

```

    is 50000
    is 135000
<Quantity 88.706 kpc>,
<Quantity 40.976 km / s>,
<Quantity 39498500.000000001 solMass>)

```